How this book is organized

The book has 11 chapters and an appendix with the following content:

- Chapter 1 provides you with an end-to-end example and defines the terminology, from sources to agents to destinations. It also discusses use cases, roles, and challenges in the context of observability.
- Chapter 2 discusses different telemetry signal types (logs, metrics, and traces), when to use which signal, how to collect signals, and the associated costs and benefits.
- Chapter 3 covers signal sources, where telemetry is generated. We discuss the types of sources that exist and when to select which source, how you can gain actionable insights from selecting the right sources for a task, and how to deal with instrumenting code you own, including supply chain aspects.
- Chapter 4 discusses different telemetry agents from log routers to OpenTelemetry. You will learn how to select and use agents, with an emphasis on what OpenTelemetry brings to the table for unified telemetry management.
- Chapter 5 focuses on backend destinations for telemetry signals, acting as the source of truth. You will learn to use and select backends for logs, metrics, and traces, with deep dives into time series databases, like Prometheus, and column-oriented datastores, such as ClickHouse.
- Chapter 6 discusses observability frontends as the place where you consume the telemetry signals. You will learn about pure frontends and all-in-ones as well as how to go about selecting them.
- Chapter 7 covers an aspect of cloud-native solutions called *cloud operations*, including how to detect when something is not working the way that it should; react to abnormal behavior; and learn from previous mistakes. You will also learn about alerting, usage, and cost tracking.
- Chapter 8 dives deep on distributed tracing and how it can help you understand and troubleshoot microservices.
- Chapter 9 dives deep into observability for developers, covering continuous profiling and developer productivity tooling.
- Chapter 10 discusses service level objectives, showing you how to use them to address the question of how satisfied the consumer of a service is.
- Chapter 11 dives deep into signal correlation, addressing the challenge of a single telemetry signal type usually not being able to answer all of your observability questions and what you can do to address this challenge.
- The appendix walks you through a complete end-to-end example, using OpenTelemetry, Prometheus, Jaeger, and Grafana.

Chapters 2 through 6 provide the conceptual foundation, so if you're entirely new to the observability space, I'd recommend working through those first. Chapters 7 through